Compiler Register Allocation

DA Programming Project II

Group T9G2 / Spring 2026

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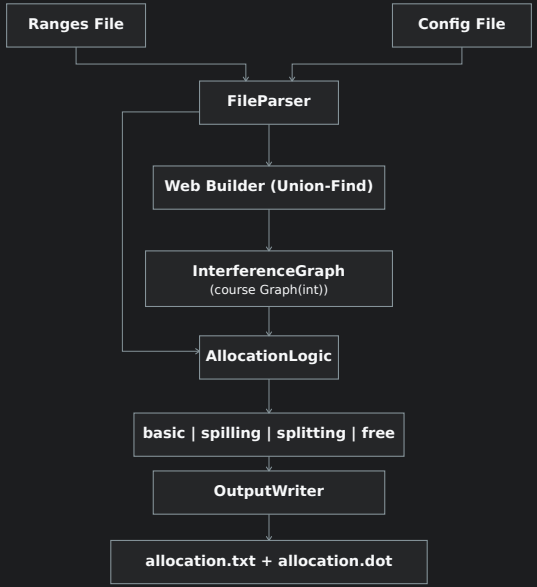
`make test` `make docs` `./register_alloc -b ...`


Ranges File
Config File
FileParser
Web Builder (Union-Find)
InterferenceGraph
(course Graph(int))
AllocationLogic
basic | spilling | splitting | free
OutputWriter
allocation.txt + allocation.dot

Input Format

Live ranges file

```
sum: 7+,8,9,10-
i: 1+,2,3,4,5,6-
i: 9+,10,11,12-
```

- var: comma-separated line numbers
- + = definition, - = last use
- same variable → multiple ranges
- # starts a comment

Register config file

```
registers: 2
algorithm: spilling, 1
```

- registers: number of physical regs
- algorithm: basic | spilling,N | splitting,N | free
- missing algorithm defaults to basic

Output Format

Allocation file

```
webs: 3
web0: 1+,2,3,4,5,6-
web1: 9+,10,11,12,14-,20+
web2: 7+,8,9,10-

registers: 2
r0: web0
r0: web2
r1: web1
```

Sections

- **Web list:** webN: sorted points
- **Assignment:** rN or M per web
- **Metadata:** spills/splits record

DOT output

- Same register = same fill color
- Memory webs → gray boxes
- Split webs → orange border
- Written alongside allocation

Basic Algorithm

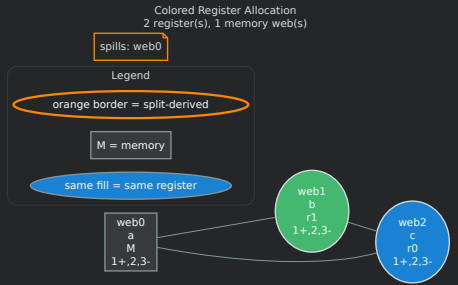
Core coloring loop

- Build interference graph from webs
- Simplify: remove node with degree < K, push to stack
- Spill candidate: select high-degree node when no low-degree node exists
- Assign: pop stack, assign available register or mark for spill
- Rewrite program and repeat until colored or bounded spilling applied

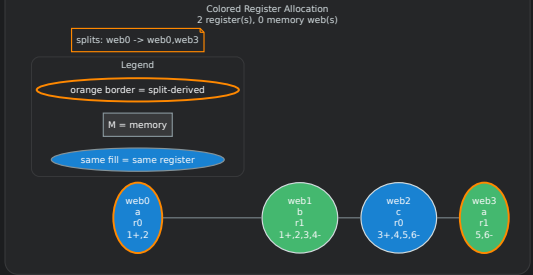
Notes: K = number of physical registers. Spill heuristic: minimize cost/degree ratio.

Spilling and Splitting

Bounded spilling



Bounded splitting

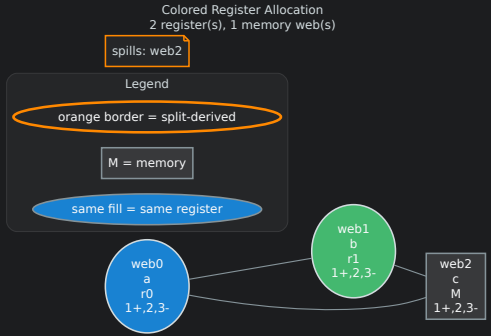


Splitting preserves original markers. 1+,2,5,6- splits into 1+,2 and 5,6- — no fabricated 2- or 5+. Metadata: spills: N / splits: N.

Custom Free Algorithm

Hybrid strategy

- Edgeless graphs: 1 register
- Complete graphs: K colored, rest spilled
- Bipartite graphs: BFS 2-coloring
- General: DSatur greedy coloring
- Small/medium: branch-and-bound refines spills



Invariant: interfering colored webs never share a register. Memory only for uncolored webs.

Demo and Testing

Inputs for the demo

- complex_allocation.txt + free4.txt: dense allocator stress test
- splitting_showcase.txt + splitting2.txt: split reduces K from 3 to 2
- spilling5.txt: bounded spilling in a dense graph

```
./register_alloc -b
  inputs/advanced/ranges/
complex_allocation.txt
  inputs/advanced/registers/free4.txt
  alloc.txt
```

make test — 76 unit + 9 integration (passes clean). DOT graphs written alongside allocation output.

Complexities

Input & graph building

Parse:	$O(LP)$
Build webs:	$O(VR^2P \log P)$
Interference:	$O(W^2P + EW)$

Coloring algorithms

Basic:	$O(W(W^2 + E))$
Spilling:	$O((S+1)W(W^2 + E))$
Splitting:	$O(SQ(W^2P + EW + W(W^2 + E)))$
Free:	$O(W^2 + E \log W)$ (heuristic)

Variables: W=webs, E=edges, P=points, R=ranges/var, S=recovery bound, Q=split candidates, L=live ranges.